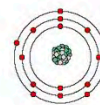


15

P

Phosphorus

Forms



15 15 16



State: Solid
Discovery: 1669

Deep phosphate mines
make up 80% of this
tiny Pacific island.



Phosphate mining in Nauru

Apatite

This is the most
common form of
the pure element.

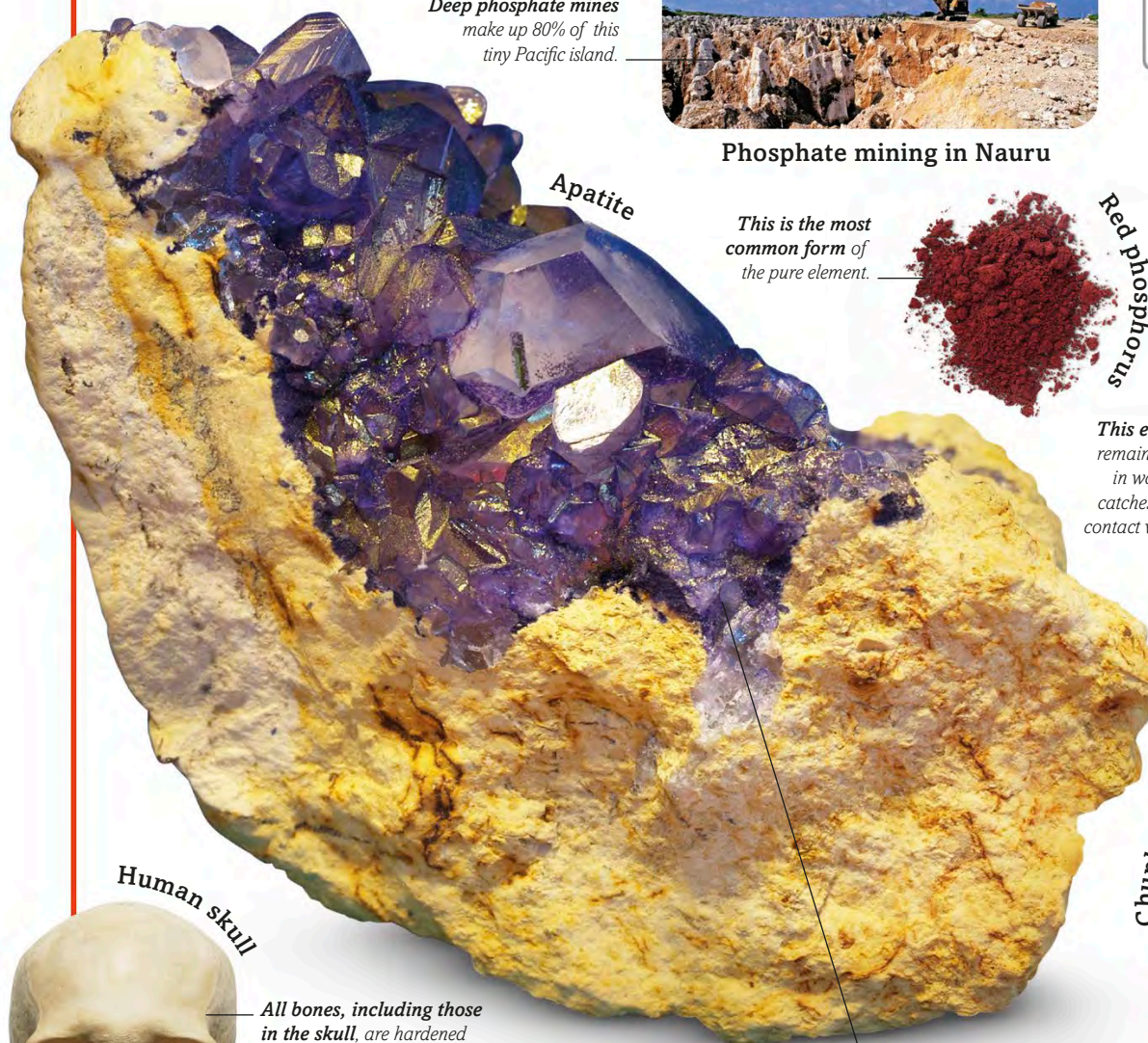


Red phosphorus

White phosphorus



This element
remains stable
in water, but
catches fire on
contact with air.



Human skull



All bones, including those
in the skull, are hardened
by calcium phosphate.

Tuna



This fish is rich
in phosphorus.

This purple colour
of the crystals is due
to metal impurities.

This form of phosphorus
is made in a laboratory by
heating red phosphorus.

Chunk of violet phosphorus



Phosphorus was accidentally discovered by German alchemist Hennig Brand. In 1669, in his quest for the mythical Philosopher's Stone (a material some believed could turn any metal into gold), he boiled a large pot full of urine for days. This produced a mysterious glowing

substance, which he called phosphorus, meaning "giver of light". Phosphorus is the first element to have a discoverer with a recorded name. It is never pure in nature, and occurs in different minerals. Phosphorus has several flammable,